

All of Us Scientific Roadmap Refresh — Executive Summary

2026 Science Committee Working Group · Co-leads: Jordan Smoller & Melissa Haendel · Draft 2026-06-02

THE QUESTION

The Science Committee asked WG-1 to refresh the All of Us Scientific Roadmap against the Roadmap 2.0 six questions: what has changed since 2023; the largest data gaps; reproducibility expectations; new tools needed; what exposures and outcomes to capture; and **whether the 8 priority focus areas are still right**. Our charge is uniquely structural — only this WG can change the framework the other four 2026 WGs must fit inside.

THE EVIDENCE BASE

We analyzed **1,428 substantive AoU publications** and **12,899 workspace projects**, theme-tagged against a locked 19-theme taxonomy. We crosswalked themes onto the 8 focus areas and computed pre/post-2023 trends. We extracted **166 strategic priorities** from **29 NIH strategic plans** (NIH-Wide + 21 ICs) and built an alignment matrix. We compared AoU against **4 peer biobanks** (UK Biobank 11,868, FinnGen 3,152, MVP 438, China Kadoorie 558 — full PubMed censuses) using a same-tagger approach. We also ran an **open-discovery horizon scan** of 294k preprints + 423k NIH grants (2021–2026) for emerging white-space. Recommendations were scored against the 8-criterion Sci Com rubric Smoller established in November 2024.

HEADLINE FINDINGS

- The **8-area framework holds at the top level** but has three structural problems: a Methods/AI orphan (18% of pubs, no home), wearables orphaned despite being a slide-7 priority, and multi-omics invisible inside “Genetics & Biology.”
- **Strongest IC alignments:** NHGRI×Genomics (largest single matrix cell; AoU delivers Bold Prediction #9), NIMHD×SDOH (Goal 7 literally describes the AoU mandate), NLM×Methods (the only IC plan that explicitly names AoU).
- **Biggest demand-side gaps:** NIEHS×Environment (6 IC priorities vs. 8 AoU pubs — the single largest gap); imaging across NCI/NEI/NIBIB (data-supply, not demand); multi-omics for NIDDK/NHLBI (the 2026 omics releases should close this mechanically).
- **Peer-biobank comparison (full-census peers):** AoU is under-indexed ~4× on rare disease, ~6× on multi-omics, ~13× vs UKB on imaging, near peer median on aging/ADRD — but **leads peers 26× on ELSI, 5× on SDOH, 4.6× on PGx, 3× on methods** (distinctive ≠ growing — AoU’s defensible identity).
- **Horizon scan — emerging white-space:** fastest-rising AoU-feasible gaps are **genomic foundation models** and **LLMs/agent AI** (AoU <0.3%; LLM-in-medicine up 47× since 2021), plus **MASLD** adoption — mapping onto the 9th focus area and AI co-scientist. Single-cell/spatial omics & AI protein structure are watch-not-build (outside AoU’s data model).
- **Researcher Workbench gaps (vs N3C, Terra/AnVIL, BioData Catalyst, UKB-RAP, Kids First; 87/87 citations source-verified):** AoU leads on data assets & GPU/AI compute, but lags on **(1)** a shared tool/workflow reuse registry (every comparator has one; 86% of AoU workspaces are solo, ~20% tutorial clones), **(2)** a governed in-environment LLM (only bring-your-own-key), and **(3)** portable-workflow/GA4GH interoperability. Top builds: governed in-VPC AI co-scientist (legal under the existing Google Cloud BAA), a reusable-apps registry, and multimodal-integration templates.
- **Pan-NIH triangulation:** five plans articulate explicit pan-IC infrastructure demand (NIH-Wide Objective 1; NLM Goal 1; NHGRI Bold Predictions #5/#9; NIMHD Goal 7; NIH-Wide CCT4/CCT5) — empirical basis for an AoU-as-pan-NIH-infrastructure framing.
- Four slide-7 priorities are **already realized** (AI 33 pubs; breakthrough therapeutics 20; wearables 75, mature; Gx×SDOH 11); four are **aspirational pending upstream investment** (multi-omics, imaging, microbiome, environmental).

TOP RECOMMENDATIONS

Recommendation	Rubric	WG / Action requested
Add a 9th focus area “ Data, AI & Methods ” (new) — absorbs the 18% Methods/AI corpus orphan, wearables, and the AI/foundation-model surge.	3.71	WG-1 — <i>endorse</i>
Rename FA8 (currently “Genetics & Biology”) to “ Genetics, Multi-omics & Biology ” and commit to a proteomics-release roadmap (Olink/SomaScan scale).	3.38	WG-1 — <i>endorse + authorize</i>
Bundle FA reframes: FA4 (“Return of Results on Individual & Population Health”) toward ROR implementation; FA2 (“Maternal & Child Health”) to “ Maternal Health ”; add explicit FA6 (“Environment”) environmental-exposure linkage commitment.	3.00	WG-1 — <i>discuss + vote</i>
AoU Workbench AI co-scientist (cohort-builder + code-generator) — highest rubric mean across all five WGs.	3.88	WG-4 — <i>fund</i>
Cross-WG bias-in-translation framework stack (WG-2 normative, WG-3 imaging, WG-4 AI) — single AoU bias-reporting standard.	3.75	WG-2 (lead) / 3 / 4 — <i>commission</i>
Plain-language distinctive-evidence narrative + annual AoU Impact Report (ELSI 26×, SDOH 5×, PGx 4.6×, methods 3×).	3.60	WG-5 — <i>fund</i>

Action key. *Endorse* — formal Sci Com expression of support (non-binding for budget). *Authorize* — permission for staff to proceed with implementation, with resource commitment. *Vote* — recorded yes/no at the meeting, required to amend the Roadmap. *Fund* — allocation of program budget in the next cycle. *Commission* — formal delivery assignment to a WG, program team, or external partner.

Current 8 focus areas. FA1 Prevalent Common & Rare Conditions · FA2 Maternal & Child Health · FA3 Healthy Aging & Resilience · FA4 Return of Results on Individual & Population Health · FA5 Lifestyle, Substance & Behavioral Health · FA6 Environment · FA7 Health Equity · FA8 Genetics & Biology.

CAVEATS

- **Dermatology over-representation (13.9% of corpus, peer index 6.9x) is an artifact** driven by 1–2 prolific groups; trend is declining (–4.4). Do not treat as a strategic priority.
- Peer-biobank counts use PubMed name-search fallback (same heuristic across peers); use the static *share index*, not trend differentials, for action-relevant comparisons.
- AoU pre-2023 baseline is tiny (157 pubs); apparent “growth” on cardiometabolic/mental health/cancer is a small-baseline artifact — read as “presence built,” not “outpacing the field.”

Full evidence base, per-IC briefs, and machine-readable companions at github.com/mellybelly/all-of-us